Background/context of the business scenario

Target audience: NHS workforce planners and Integrated Care Board decision makers assessing whether rising appointment demand necessitates structural capacity expansion or can be managed through targeted optimisation of existing General Practice resources.

Stakeholders disagree on whether expansion is required or whether better utilisation of existing resources would be adequate. Rising population, missed appointments, and changing service delivery models increase financial and operational uncertainty. The business problem is therefore to assess historical utilisation trends, appointment volumes, duration patterns, and missed appointment behaviour to evaluate whether current capacity has been adequate and to identify recommendations for future resource allocation and service optimisation.

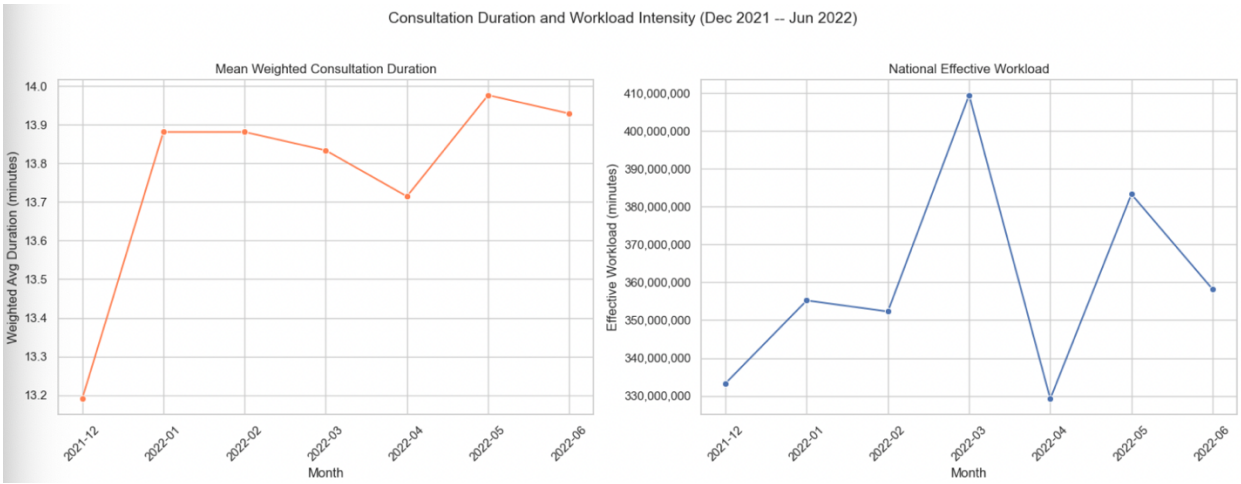
Analytical approach

NHS datasets were loaded using pandas: pd.read_csv() for actual_duration.csv and appointments_regional.csv, and pd.read_excel() for national_categories.xlsx. tweets.csv was imported to analyse public sentiment through hashtag frequency.

Data inspection used .info(), .describe(), .isna().sum(), and .shape to confirm structure and completeness. No critical missing values were identified. Metadata highlighted a known quality limitation: appointment recording lacks national standardisation, so fields such as appointment_mode and hcp_type may vary in accuracy across regions. A key constraint was a temporal mismatch. actual_duration covers December 2021 to June 2022, while appointments_regional spans January 2020 onwards. Duration based analysis was therefore limited to seven months. The datasets were merged at ICB month level using a left join to preserve the full appointment timeline while attaching duration data where available. Date fields were converted with pd.to_datetime() and standardised to monthly periods using .dt.to_period("M"). Duration bands such as "6 to 10 Minutes" were mapped to numeric midpoints to compute weighted average consultation duration per ICB per month.

Business questions were structured around themes: service utilisation, missed appointments, demand versus capacity. Targeted groupby() aggregations were applied across service_setting, context_type, national_category, hcp_type, appointment_status, appointment_mode, and time_between_book_and_appointment. Derived metrics included a daily utilisation proxy calculated as monthly totals divided by 30, a DNA rate calculated as DNA divided by total appointments multiplied by 100, effective workload in minutes, and a duration coverage rate measuring the proportion of appointments with usable duration data. Seasonal subsets were created by mapping appointment_month to season labels.

To explore whether expansion was required or if demand could be met through better use of existing resources, expansion was defined as: hiring more GPs, increasing clinic hours, etc. The appointments_regional dataset was merged with actual_duration, left on icb_ons_code to calculate the weighted average duration of appointments and compare it to national effective workload. The result was a peak in workload in March 2022, suggesting that targeted seasonal capacity planning and improved utilisation of existing resources may be more effective immediately, before considering structural expansion in the long term.



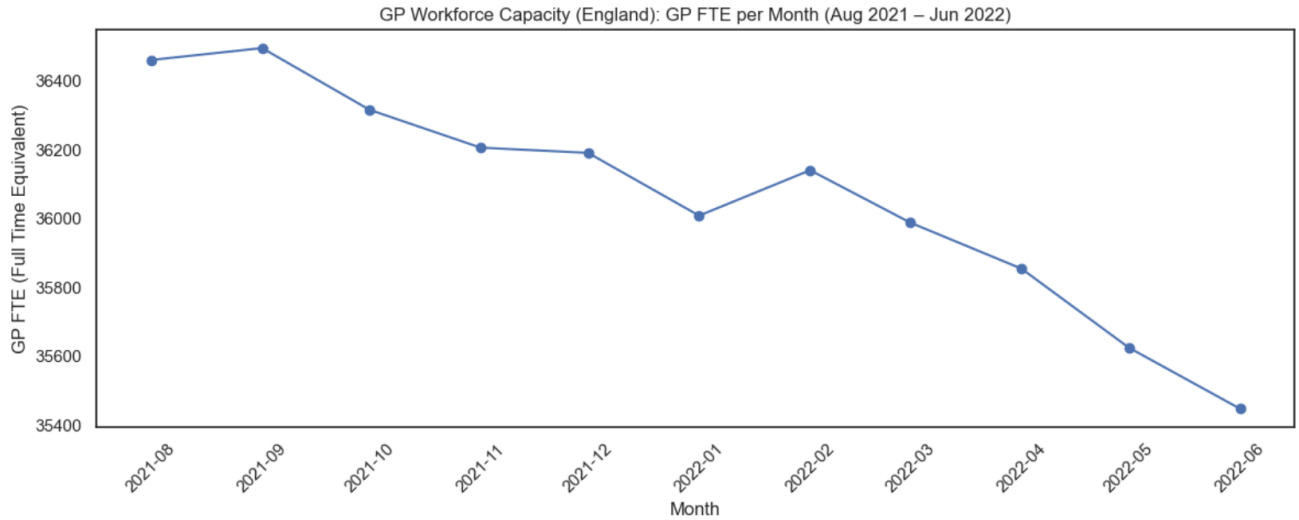
Visualisation and insights

Line charts were used first to explore monthly appointment volumes across service settings, context types and national categories. These charts show clear seasonality, with peaks in autumn and March and dips in December and April. General Practice consistently accounts for the majority of activity. Because GP volumes are significantly larger than other service settings, separate subplots were created to maintain readability; combining them on a single scale would visually compress smaller categories and reduce interpretability.

Seasonal bar charts were then used to communicate the system-level dependence on General Practice in a clearer business format. These explanatory visualisations simplify the time-series detail and highlight that alternative service settings contribute relatively small volumes compared to core GP services.

Workforce capacity was incorporated using external GP Full-Time Equivalent (FTE) data. FTE declined by approximately 2.78% over the period. To meaningfully compare demand and capacity, appointments were normalised by FTE to calculate appointments per GP.

This workload metric shows seasonal peaks rather than a sustained upward trend. Demand fluctuates, but there is no clear structural acceleration in workload per GP within the 11-month window.

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Percentage decrease in GP FTE (Aug 2021 → Jun 2022): 2.78%
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99]:

	metric	month	fte
0	All GPs 10,11,13	2021-08-01	36,461
1	All GPs 10,11,13	2021-09-01	36,495
2	All GPs 10,11,13	2021-10-01	36,316
3	All GPs 10,11,13	2021-11-01	36,206
4	All GPs 10,11,13	2021-12-01	36,191
5	All GPs 10,11,13	2022-01-01	36,009
6	All GPs 10,11,13	2022-02-01	36,141
7	All GPs 10,11,13	2022-03-01	35,988
8	All GPs 10,11,13	2022-04-01	35,855
9	All GPs 10,11,13	2022-05-01	35,626
10	All GPs 10,11,13	2022-06-01	35,449

Figure: Gradual 2.78 Percent Decline in GP FTE Workforce from August 2021 to June 2022 Indicating Structural Capacity Pressure Rather Than Short Term Fluctuation (explanatory – in a demand vs capacity analysis, this chart illustrates the capacity)

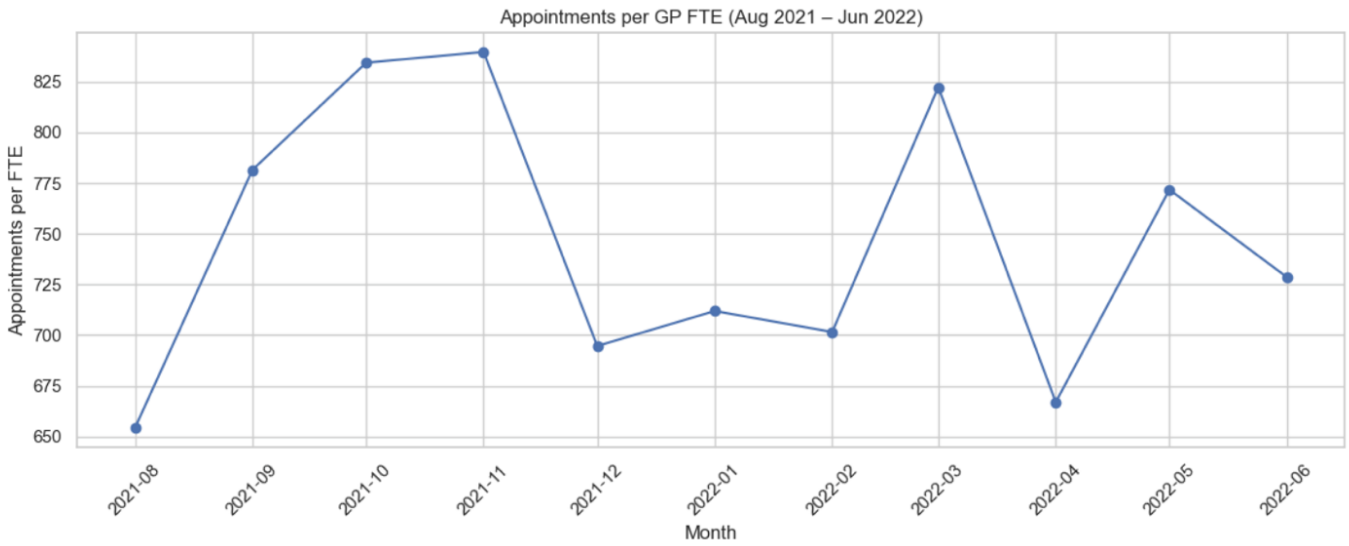

Figure: seasonal spikes in the ratio of appointments per GP FTE highlighting fluctuating workload intensity despite relatively stable workforce levels (explanatory- evaluating operational pressure by showing how workload per GP changes over time)

Limitations include the short timeframe (August 2021 – June 2022), national-level aggregation which may mask regional pressure, absence of case complexity data, and the assumption that each FTE contributes uniformly to appointment delivery. The dominance of General Practice in total volumes may also obscure pressures in smaller but critical service areas.

Overall, the evidence suggests cyclical operational pressure rather than immediate structural overload, supporting targeted seasonal planning and workforce retention rather than blanket expansion.

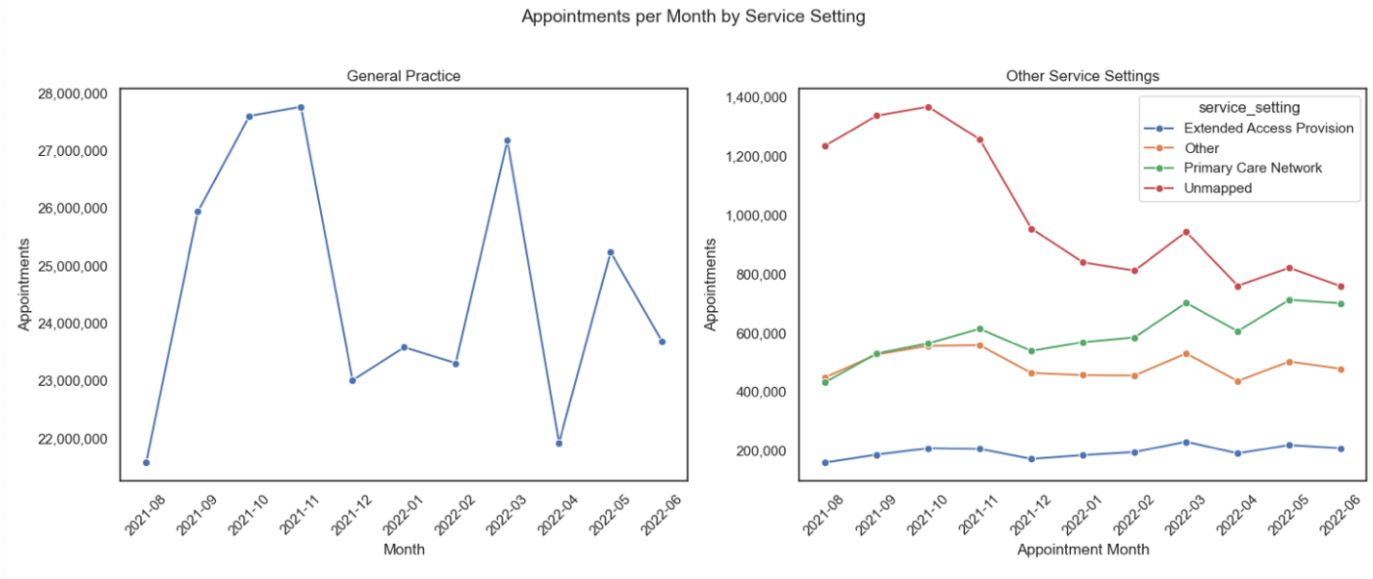

Figure: Seasonal Fluctuations Observed Across NHS Services, with General Practice Maintaining the Highest Utilisation Throughout (exploratory)

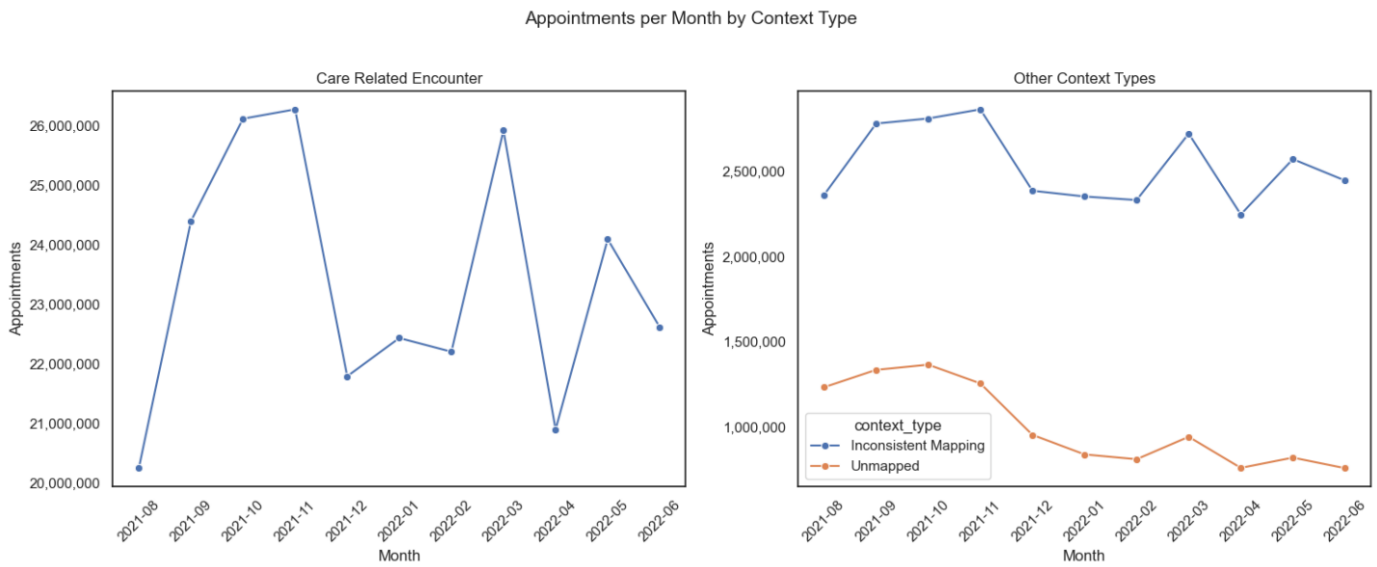

Figure: Care Related Encounters Drive Appointment Volumes While Unmapped Cases Decline Over Time (exploratory)

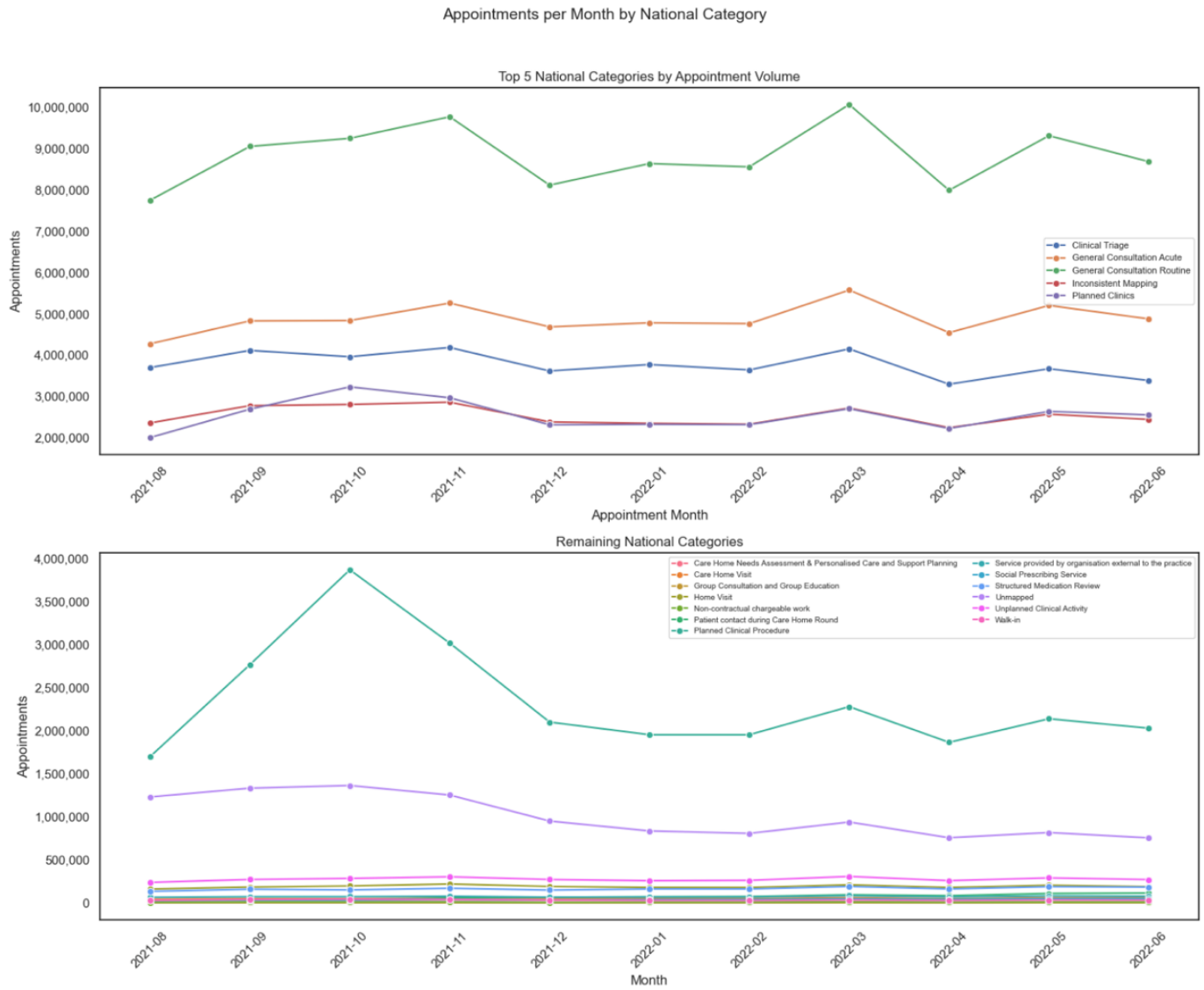

Figure: General Consultation Categories Drive NHS Appointment Volume with Clear Seasonal Fluctuations Across All Categories (exploratory- visualises overall distribution and seasonality across appointment types)

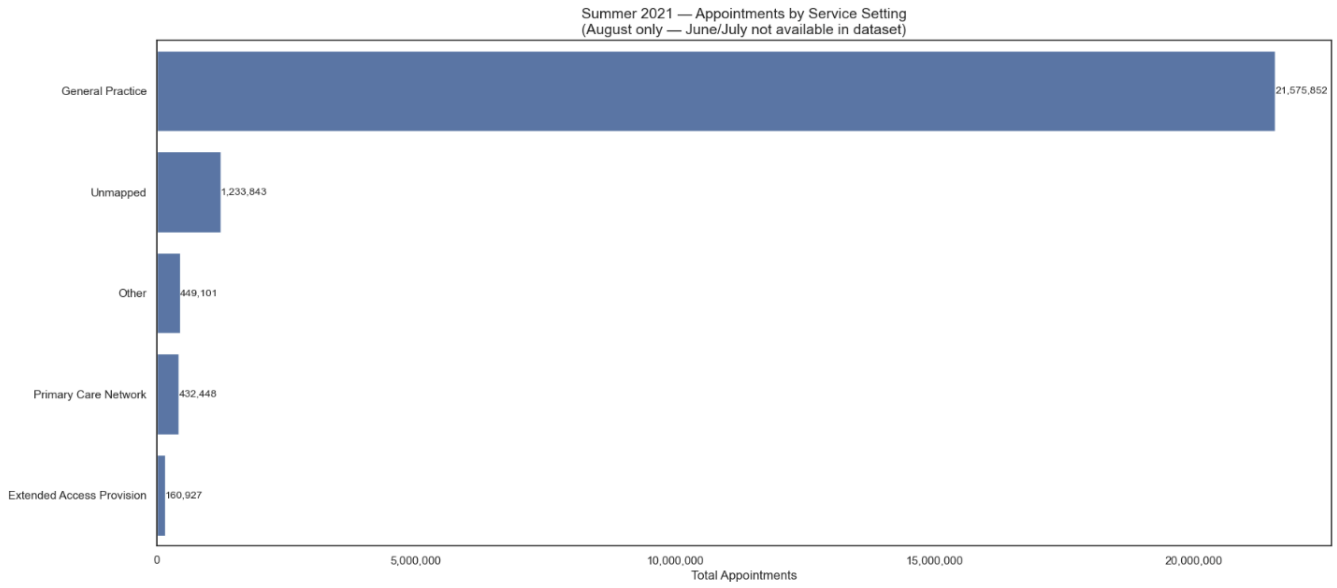

Figure: General Practice Accounts for the Vast Majority of NHS Appointments in Summer 2021 (explanatory- GP services drive the most appointments in each season)

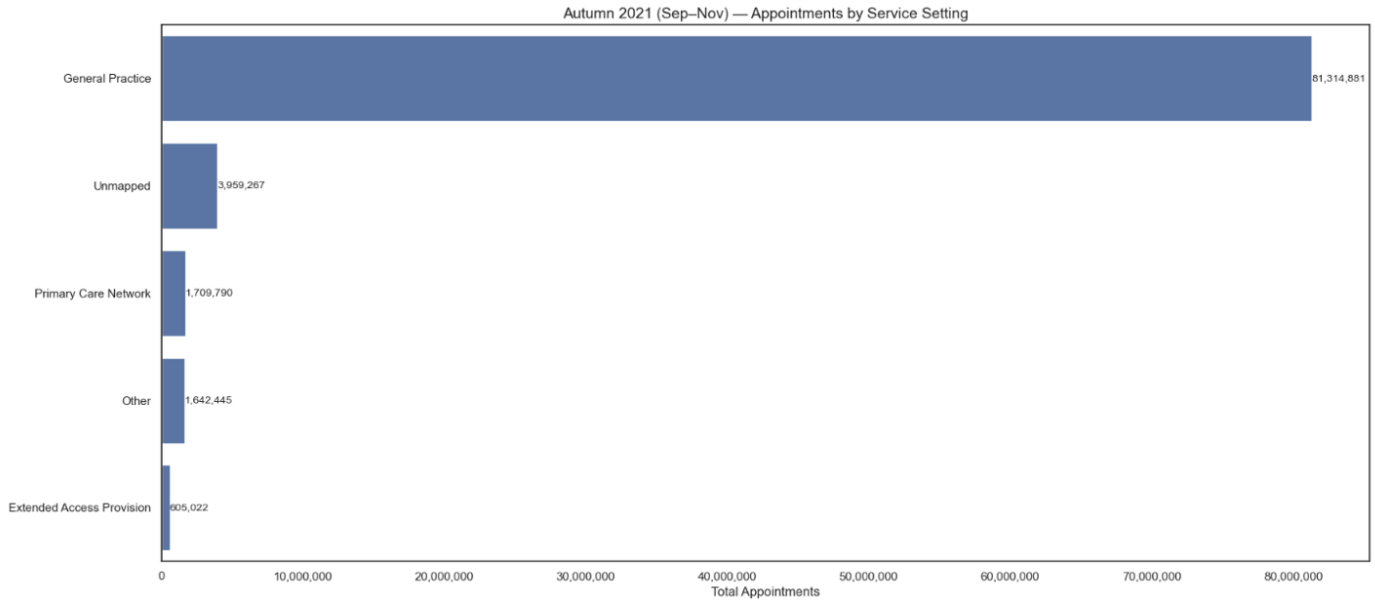

Figure: Quarterly Appointment Volumes in Autumn 2021 Demonstrate System-Level Dependence on General Practice, with Alternative Service Settings Playing a Minor Role (explanatory)

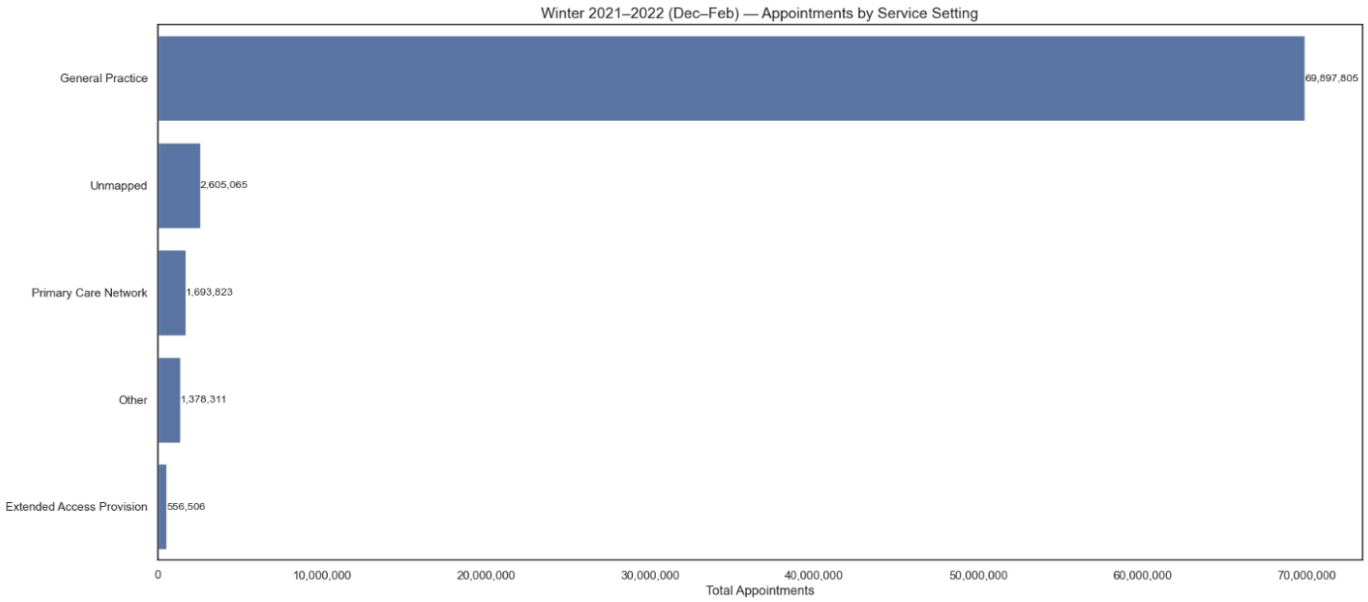

Figure: Winter 2021–2022 Appointment Volumes Confirm Continued Dominance of General Practice (explanatory)

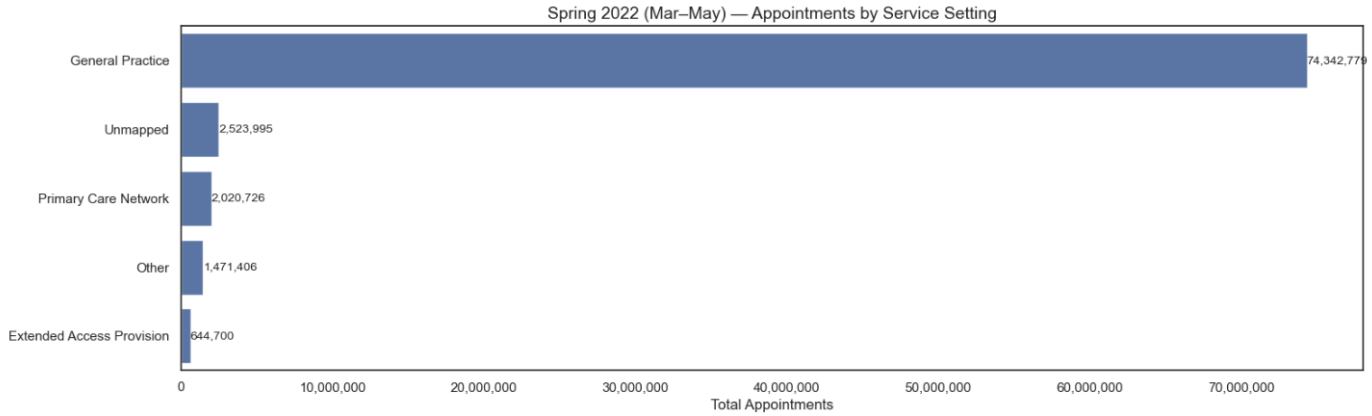

Figure: Spring 2022 Shows Rising Appointment Volumes but Continued Concentration in General Practice (explanatory- demand growth remains in GP services)

	month	total_appts	dna_appts	dna_rate_percent
0	2020-01-01	27199296	1298269	5
1	2020-02-01	24104621	1215154	5
2	2020-03-01	24053468	1166314	5
3	2020-04-01	16007881	478766	3
4	2020-05-01	16417212	449057	3
5	2020-06-01	20690805	594382	3
6	2020-07-01	22491437	698327	3
7	2020-08-01	20150520	669462	3
8	2020-09-01	26714255	1145971	4
9	2020-10-01	28301932	1358138	5
10	2020-11-01	25061602	1021408	4
11	2020-12-01	23535936	996416	4
12	2021-01-01	22492069	899833	4
13	2021-02-01	22399569	804659	4
14	2021-03-01	27225424	964719	4
15	2021-04-01	23879932	893723	4
16	2021-05-01	23508395	878219	4
17	2021-06-01	26784182	1030367	4
18	2021-07-01	25739219	1029665	4
19	2021-08-01	23852171	949137	4
20	2021-09-01	28522501	1321348	5
21	2021-10-01	30303834	1565624	5
22	2021-11-01	30405070	1428087	5
23	2021-12-01	25140776	1198866	5
24	2022-01-01	25635474	1076013	4
25	2022-02-01	25355260	1076658	4

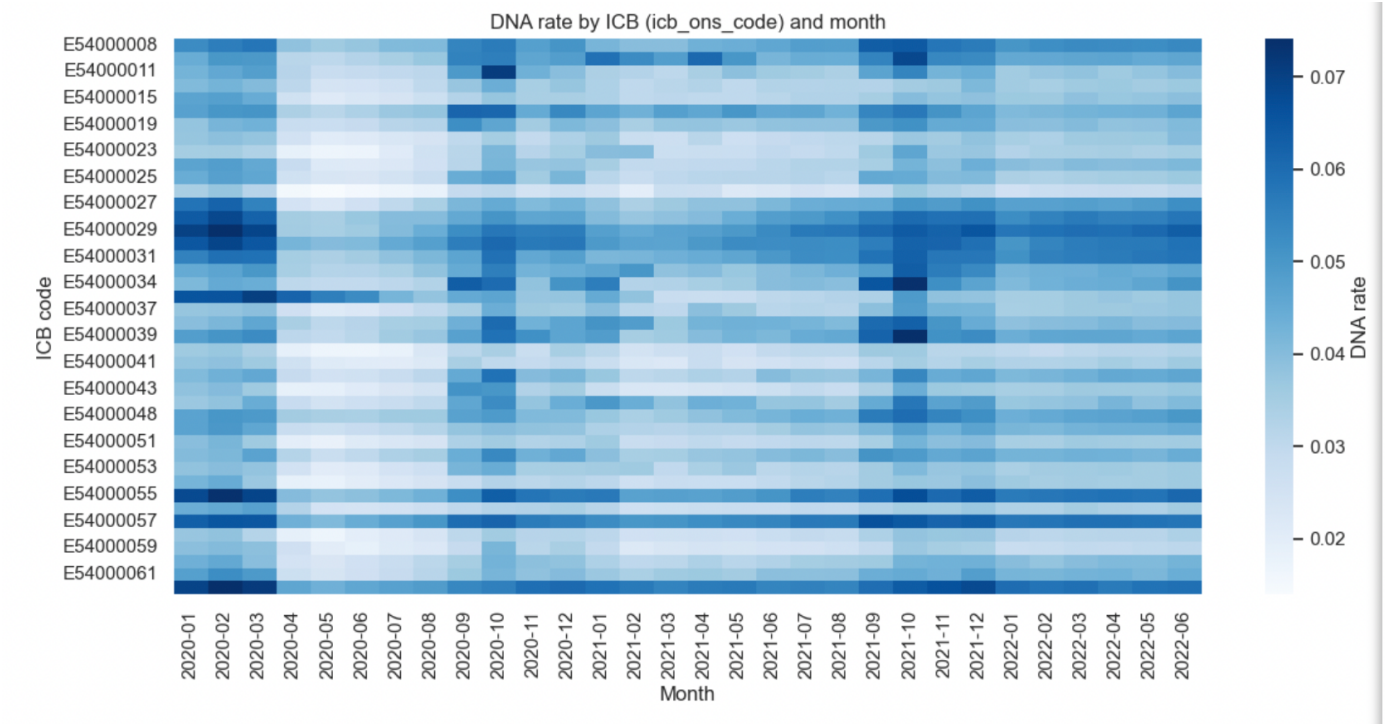

Figure: heatmap showing variation in DNA rates across ICBs over time, highlighting regional differences and seasonal patterns in missed appointments (exploratory- it visually scans for patterns, clusters, and anomalies across regions and months without assuming a cause for patterns)

Patterns and predictions

Monthly NHS appointments show a pronounced rise from August to October–November 2021, followed by a sharp decline in December, likely reflecting seasonal effects. Activity recovers in March 2022 before falling again in April, indicating recurring volatility rather than sustained growth. General Practice consistently accounts for the vast majority of total appointments, demonstrating structural dependence on GP services across the system. Other service settings exhibit similar directional patterns but at significantly lower volumes. While GP FTE declines modestly over the period, appointment intensity per GP fluctuates, suggesting that short-term workload pressures are driven primarily by seasonal demand variation rather than immediate workforce expansion needs.

Areas of further exploration include incorporating sheet 1b of the GPW Bulletin Tables – June 2022 Excel workbook. This is because sheet 1b includes GP headcount by GP role type (Partners, Salaried, Locums, Trainees) which would allow comparison between total FTE and total headcount over time. Such a comparison would enable calculation of average FTE per GP, providing insight into whether changes in capacity are driven by fewer

clinicians overall or by reductions in average working hours. If headcount remains relatively stable while FTE declines, this may indicate a shift towards part time working patterns rather than a recruitment shortfall.